

FitFlex

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Contents

Chapter 1 - Introduction	1
1.1 Problem Statement	1
1.2 Scope	1
1.3 Modules	2
1.3.1 Module 1 - Real Time Workout Analysis	2
1.3.2 Module 2 - Accessory Store	2
1.3.3 Module 3 - Community and Forums	2
1.3.4 Module 4 - Find Nearby Gyms	3
1.3.5 Module 5 - Fitness Chatbot	3
1.3.6 Module 6 - Personalized Workout	3
1.3.7 Module 7 - Mobile Application	3
1.3.8 Module 8 - Performance and Progress Reports	4
1.4 User Classes and Characteristics	4
Chapter 2 - Project Requirements	5
2.1 Use-case Diagram	5
2.2 Functional Requirements	7
2.3 Non-Functional Requirements	9
2.4 Domain Model	10
Chapter 3 - System Overview	11
3.1 Architectural Design	11
3.2 Design Models	12
Activity Diagram:	12
Class Diagram:	13
Class Level Sequence Diagrams:	13
State Transition Diagram	18
System Sequence Diagrams	19
3.3 Data Design	24
ERD	25
Bibliography	26

List of Figures

Figure 1 use case Diagram.....	2
Figure 2 Domain model.....	9
Figure 3 Architecture diagram	10
Figure 4 Activity Diagram.....	12
Figure 5 Class Diagram	13
Figure 6 Sequence Diagram : Chatbot	13
Figure 7 Sequence Diagram : Create post	13
Figure 8 Sequence Diagram : Real-time workout Analysis	13
Figure 9 Sequence Diagram : Find nearby Gyms.....	13
Figure 10 Sequence Diagram : Nearby Gyms	13
Figure 11 Sequence Diagram : View Posts	13
Figure 12 Sequence Diagram : Gym Accessories	13
Figure 13 state transition diagram	13
Figure 14 System Sequence Diagram : View accessories	13
Figure 15 System Sequence Diagram : community forums.....	13
Figure 16 System Sequence Diagram : Personalised Workout	13
Figure 17 System Sequence Diagram : Real time Workout Analysis	13
Figure 18 System Sequence Diagram : View Nearby Gyms	13
Figure 19 Entity Relationship Diagram	13

Chapter 1 - Introduction

1.1 Problem Statement

While being fit and staying in shape is the dream of many, achieving it might not actually be as simple as it sounds. A core component of performing an exercise is its correct form. A bad exercise form can not only impact fitness growth but can also result in injuries. According to a recent survey, approximately 500,000 people get injured in the gym annually only in the US. A big percentage of these injuries are due to improper exercise form and lack of professional guidance. In current times, we see many different fitness gurus and coaches who themselves are very less aware of the correct exercise forms. With these local gym coaches and social media fitness mentors flooding the fitness world, authenticating their credibility still remains a challenge. This is resulting in a large majority of people getting injuries and not being able to reach their fitness goals. Such people then become demotivated and then continue to neglect their fitness throughout their lives.

1.2 Scope

The project is an AI based fitness application which will function as a virtual Gym trainer.. Below are the features we wish to implement:

- Provide users with real time audio feedback on their exercise form through the phone camera
- Provide personalized workout plans tailored to the users requirement such as losing weight,bulking,cardio etc.
- Show the user gyms near him along with their contact information.
- Present the user with the best deals on Gym equipment and fitness related products

- Count reps in real time .
- Chatbots for gym/workout have a Gym community and forums .
- Generate performance reports

Having implemented these features, our application will prove to be a one stop solution for all fitness needs.

1.3 Modules

1.3.1 Module 1 - Real Time Workout Analysis

Fitflex will perform real time workout analysis through computer vision. The application will analyze the user's range of motion and posture of exercise in real time and will help correct the form. Features of this module are:

1. Real time audio feedback
2. Real time rep counter
3. Generate performance reports

1.3.2 Module 2 - Accessory Store

The application will include an accessory store which will provide a collection of gym accessories with their prices. Users will be provided with the best deals on Gym accessories from major stores in Pakistan. Following are some of the features included in this module:

1. Search filters based on type of accessories, brands etc.
2. Product comparison

1.3.3 Module 3 - Community and Forums

Application will have a community and forums module for enhanced user interaction. Users will be able to interact with each other to share their experiences, ask questions and provide feedback and recommendations. Following are some of the features of this module:

1. Create posts
2. Like, reply and react to posts

1.3.4 Module 4 - Find Nearby Gyms

This module will help users to find nearby gyms from their location. Users will be able to find and search for gyms based on distance. Features of this module include

1. View contact information of nearby Gyms
2. Check timings
3. Like and save gyms
4. Get directions

1.3.5 Module 5 - Fitness Chatbot

A fitness chatbot will be available to answer user's basic queries. The chatbot will help the user with search for exercises, injury information, etc. The features include:

1. Search for exercises based on muscle groups, level of expertise and equipment availability.
2. Offer injury prevention and treatment recommendations
3. Help the user with their diet goals

1.3.6 Module 6 - Personalized Workout

Different users have different workout goals. Our personalized workouts will help the user achieve their unique fitness goals. Whether it is getting lean, increasing muscle mass, enhancing endurance, or powerlifting, this module will have everything covered. The main features of this module are:

1. Customized workout plans based on fitness goals
2. Updating plans based on user feedback

1.3.7 Module 7 - Mobile Application

This module covers the application development for the project. A versatile application with user-friendly frontend and robust backend will be developed. Key aspects of this module will include:

1. A seamless front-end for the users to make navigation easier
2. A robust and scalable backend for efficient data handling and application functionality
3. A feedback mechanism will allow the users to report any bugs and glitches in the app
4. Multiple theme options will be available

1.3.8 Module 8 - Performance and Progress Reports

In this module, performance reports will be generated by analyzing the user's workouts. These reports will then be used to identify the user's progress to generate recommendations. Key aspects of this module include:

1. Tracking records of recommendations generated.
2. Progress reports to identify progress made and weaknesses to be worked on.

1.4 User Classes and Characteristics

User Class	Description
Fitness Enthusiasts - 15-45 years	This is the primary user class which will benefit from real time workout monitoring, personalized workout plans and use the chatbot for fitness related guidance.
Fitness Community Members	These are the users who like to share their fitness experiences with fellow members in a community or Forum. Members of this user class will use the application to share their fitness journeys by creating posts, and engaging with fellow forum members on their posts.
Gym Accessory Purchasers	This group of users will be looking for fitness related accessories. Members of this user class can use the application to browse for gym accessories and purchase them.

Chapter 2 - Project Requirements

2.1 Use-case Diagram

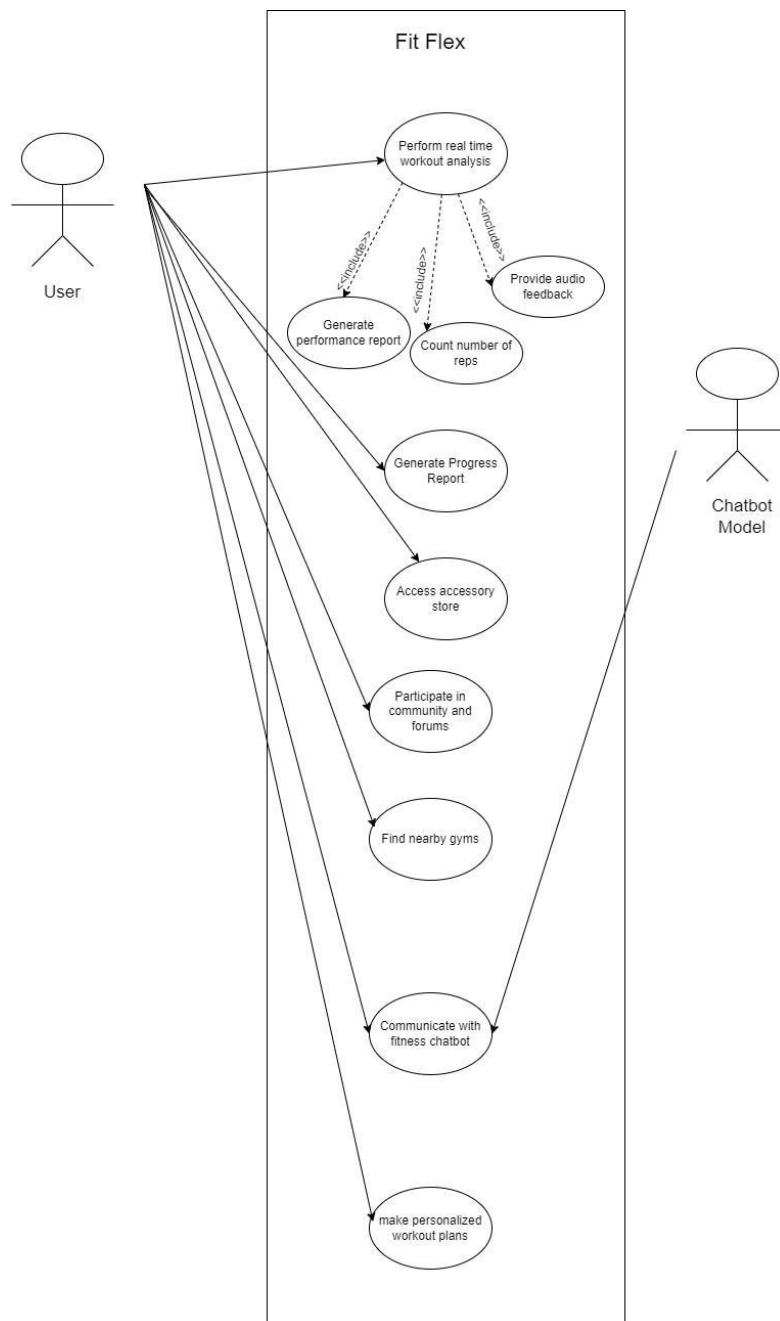


Figure 1 use case Diagram

2.2 Functional Requirements

Below mentioned are the functional requirements for each module:

2.2.1 Module 1 - Real Time Monitoring

- **FR1** : System shall capture real time video of the user through the phone camera.
- **FR2** : System shall analyze the user's video for form correction.
- **FR3** : System shall provide audio feedback to the user in case of wrong exercise form.
- **FR4** : System shall count exercise repetitions in real time.
- **FR5** : System shall generate performance reports for the exercise including number of reps and form correctness.

2.2.2 Module 2 - Accessory Store

- **FR1** : System shall display gym accessories with their prices.
- **FR2** : System shall provide search filters to users.

2.2.3 Module 3 - Community and Forums

- **FR1** : System shall allow users to create posts
- **FR2** : System shall allow users to react and reply to other users' posts

2.2.4 Module 4 - Find Nearby Gyms

- **FR1** : System shall display nearby gyms to users based on their current location.
- **FR2** : System shall display contact details for each gym
- **FR3** : System shall allow the users to like and save gyms for future reference.

2.2.5 Module 5 - Fitness Chatbot

- **FR1** : System shall allow the users to find exercises based on muscle groups through chatbot interaction.
- **FR2** : The chatbot shall offer fitness related guidance to users like correct form and technique and injury prevention tips.
- **FR3** : System shall provide nutritional advice to users and help them achieve their diet goals.

2.2.6 Module 6 - Personalized Workout

- **FR1** : System shall allow users to add fitness goals and current fitness level.
- **FR2** : System shall provide personalized fitness plans to users based on their fitness goals.

2.2.7 Module 7- Mobile Application

- **FR1** : The application shall allow the users to select different themes for customization.
- **FR2** : System shall provide a feedback mechanism to report for bugs and errors in the application.

2.2.8 Module 8- Progress Reports

- **FR1** : The system shall keep track of performance reports per exercise.
- **FR2** : System shall indicate progress points and weak points through the report.

2.3 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.3.1. Reliability:

- The system shall have an MTBF (Mean Time Between Failures) of at least 1,000 hours for all major functionalities, including workout analysis, chatbot responses, and personalized workout recommendations.
- In the event of a failure, the system shall be able to recover to a stable state within 5 minutes without data loss.
- The application shall have an uptime of at least 90% over any 30-day period, excluding scheduled maintenance windows
- the application shall be able to handle at least 100 concurrent users without degradation in performance or functionality.

2.3.2. Usability:

- The system shall have an intuitive user interface with a learning curve of no more than 2 interactions for any given feature (e.g., searching for a gym, viewing performance reports).
- The application shall support text scaling up to 200% without loss of content or functionality to accommodate users with visual impairments.
- First-time users shall be able to set up their profile and start their first workout within 5 minutes

2.3.3. Performance:

- The system shall process and provide real-time audio feedback for workout

corrections with a response time of less than 1 second.

- The system shall display real-time workout data (e.g., rep counts, form analysis) within 2 seconds of receiving sensor input.
- The system shall retrieve and display personalized workout plans within 3 seconds of a user request.
- Pose estimation and form evaluation shall process and provide feedback within 1 second of receiving a video frame.
- Find nearby Gyms shall return a list of nearby gyms within 2 seconds for any **given location**.

2.3.4 Security

- The application shall enforce a password policy requiring at least 10 characters, including uppercase and lowercase letters, numbers, and special characters.
- The system shall implement role-based access control (RBAC) to prevent unauthorized access to user accounts
- All passwords and personal data shall be encrypted using hashing algorithms
- The system will use JWT tokens to handle user authentication

2.4 Domain Model

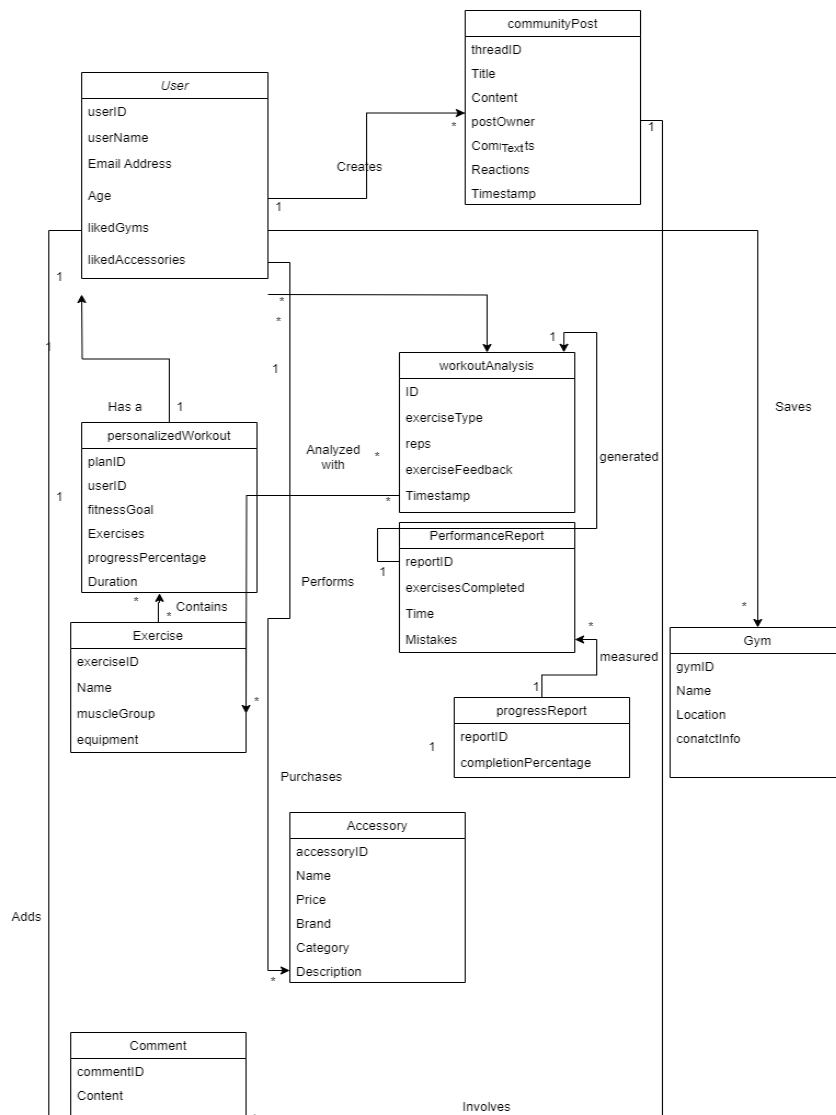


Figure 2 Domain model

Chapter 3 - System Overview

3.1 Architectural Design

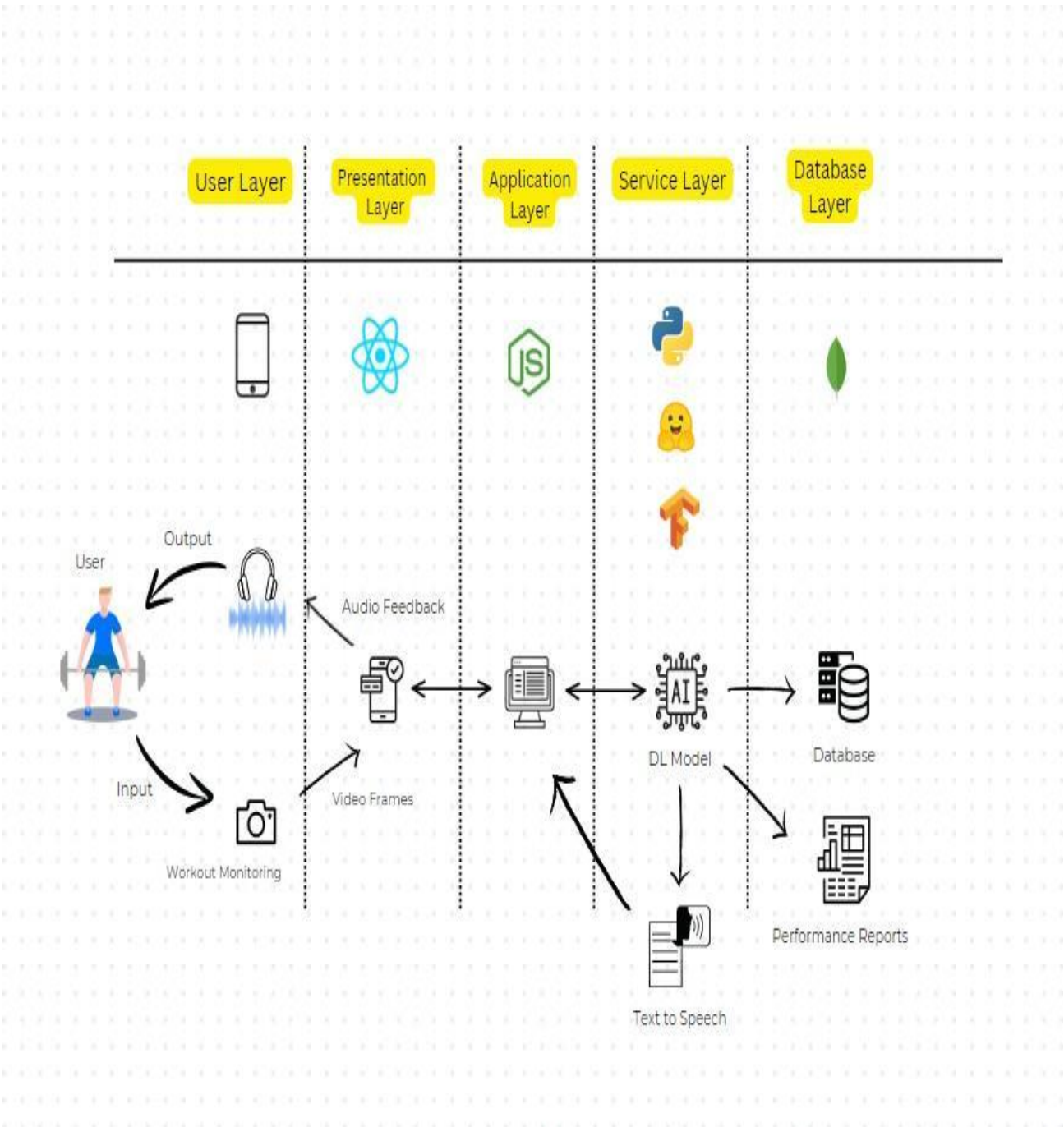


Figure 3 Architecture diagram

3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

Design Models for Object Oriented Development Approach

The applicable models for the project using object oriented development approach may include:

- 3 Activity Diagram
- 4 Class Diagram
- 5 Class-level Sequence Diagram
- 6 State Transition Diagram (for the projects which include event handling and backend processes)

Design Models for Procedural Approach

The applicable models for the project using procedural approach may include:

- 7 Activity Diagram
- 8 Data Flow Diagram (data flow diagram should be extended to 2-3 levels. It should clearly list all processes, their sources/sinks and data stores.)
- 9 System-level Sequence Diagram
- 10 State Transition Diagram (for the projects which include event handling and backend processes)

Approach : Object Oriented Development

Activity Diagram:

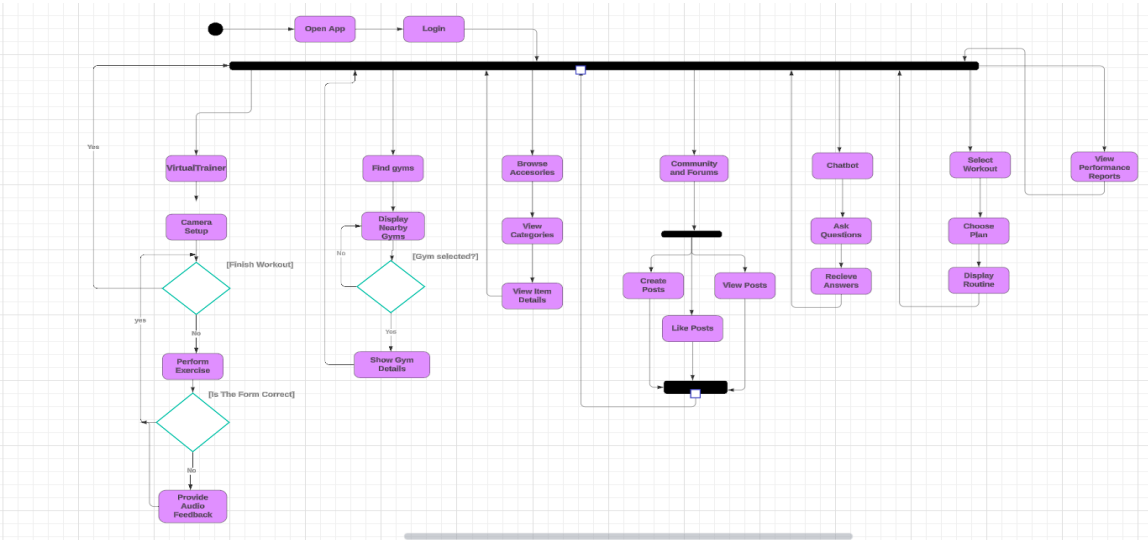


Figure 4 Activity Diagram

Class Diagram:

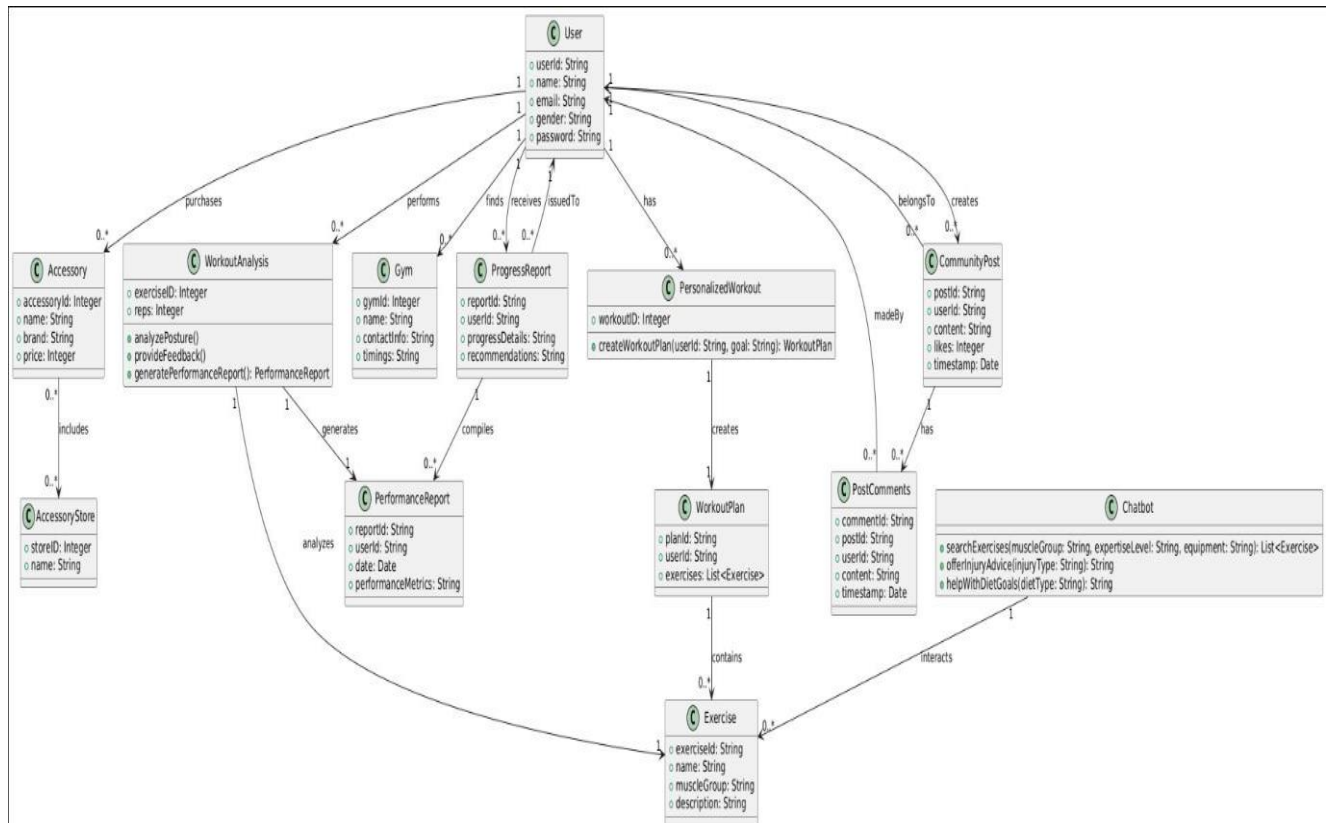


Figure 5 Class Diagram

Class Level Sequence Diagrams:

1) Chatbot:

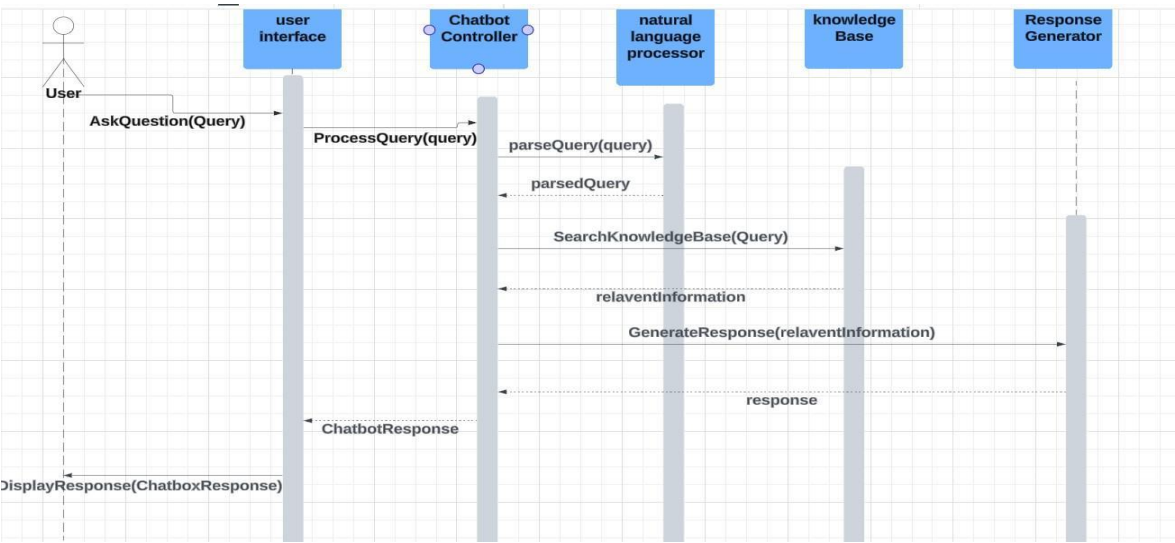


Figure 6 Sequence Diagram : Chatbot

2) Create Posts

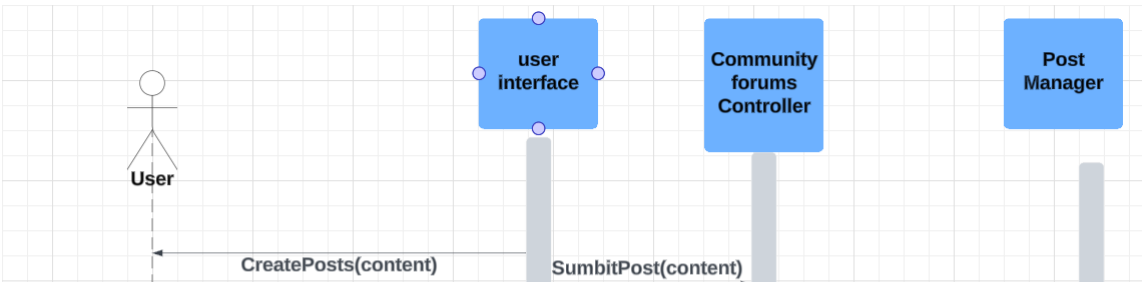


Figure 7 Sequence Diagram : Create post

3) Real-Time Workout Analysis

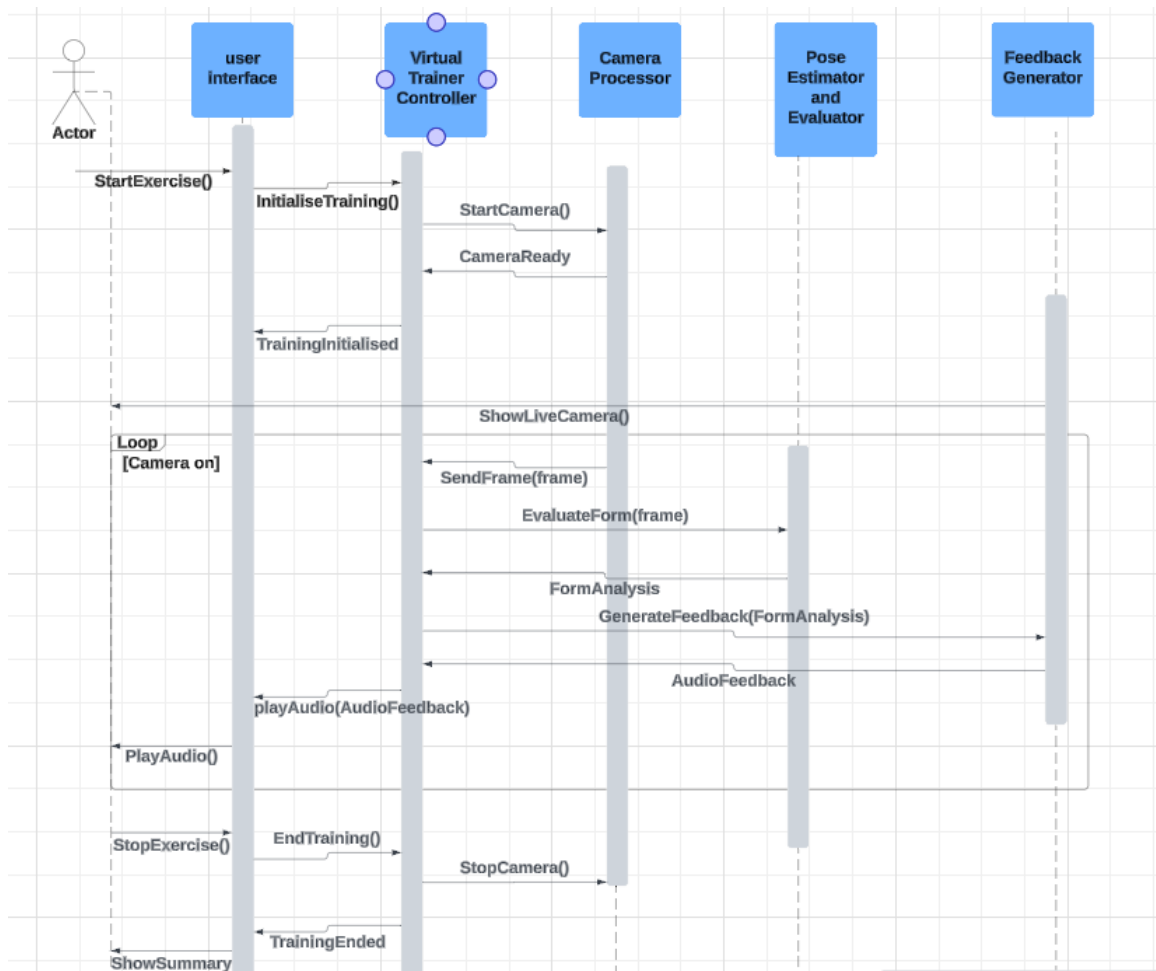


Figure 8 Sequence Diagram : Real-time workout Analysis

4) Find Nearby Gyms

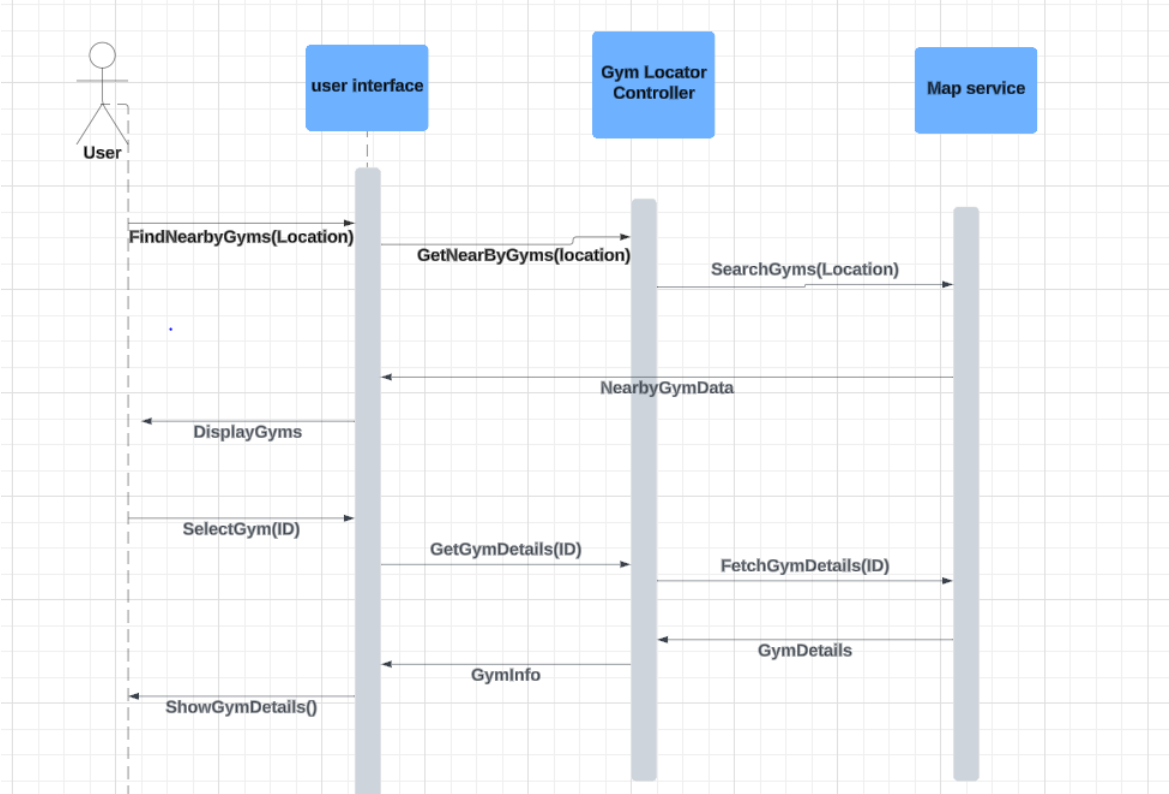


Figure 9 Sequence Diagram : Find nearby Gyms

5) Personalized Workout

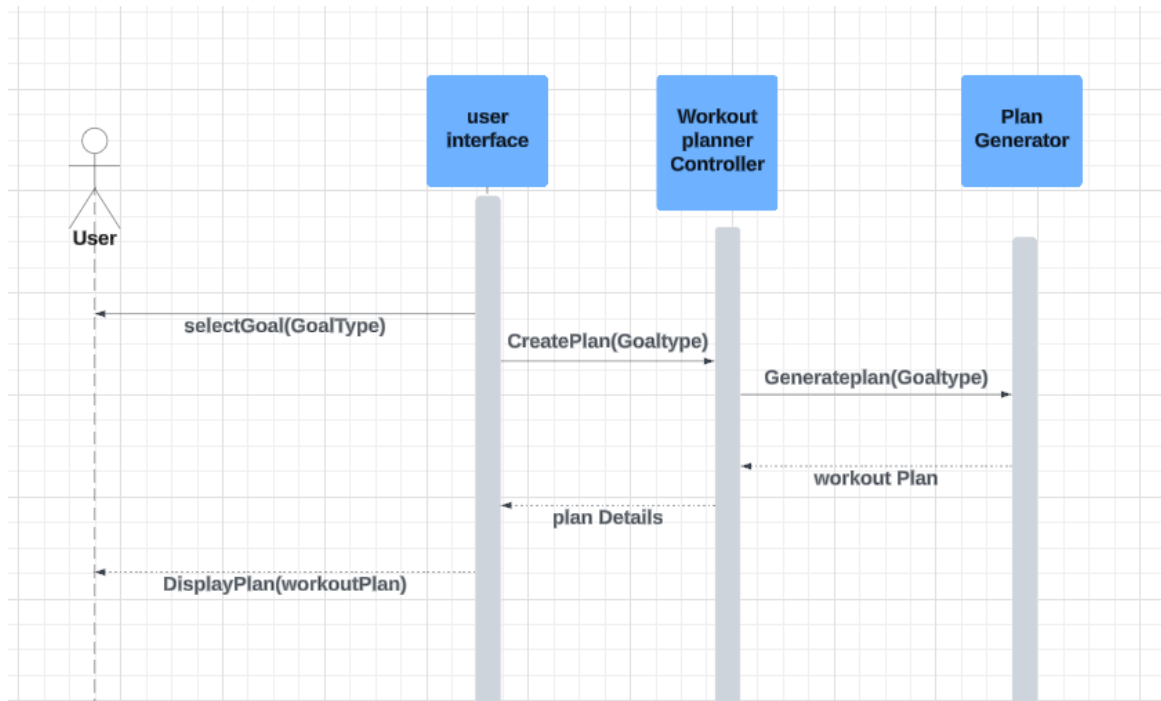


Figure 10 Sequence Diagram : Nearby Gyms

6) View Posts

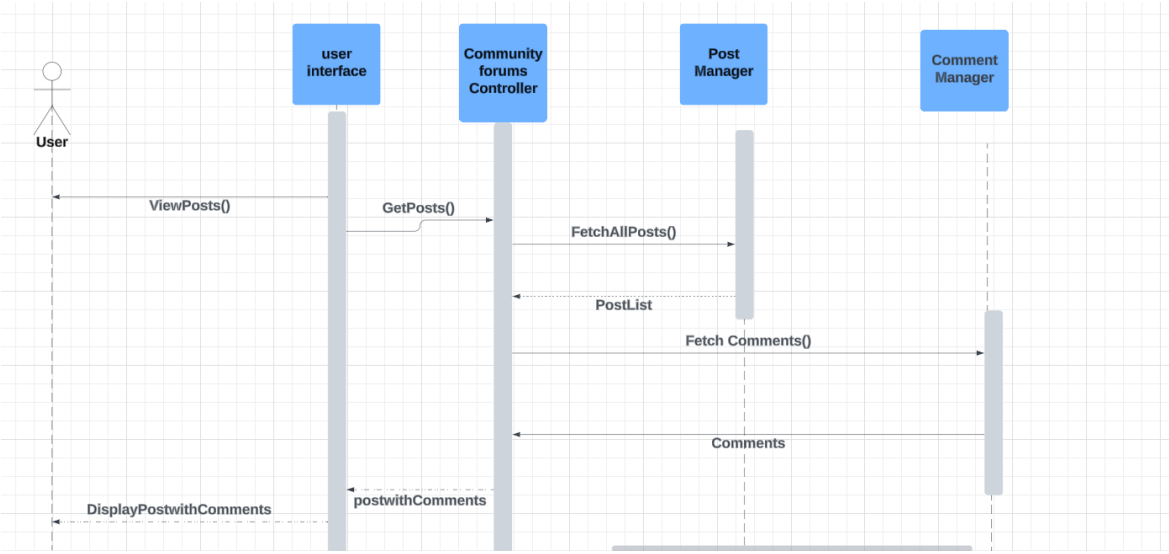


Figure 11 Sequence Diagram : View Posts

7) Gym accessories

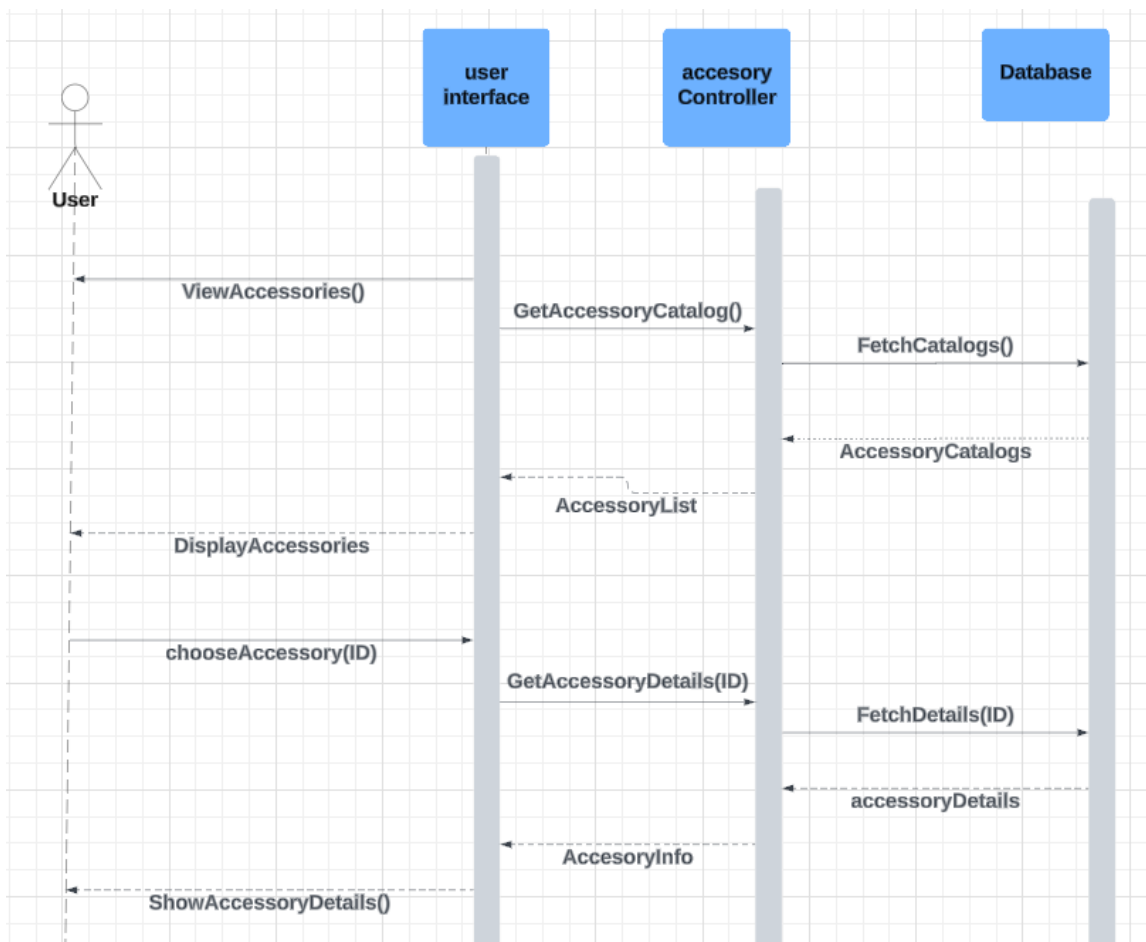


Figure 12 Sequence Diagram : Gym Accessories

State Transition Diagram

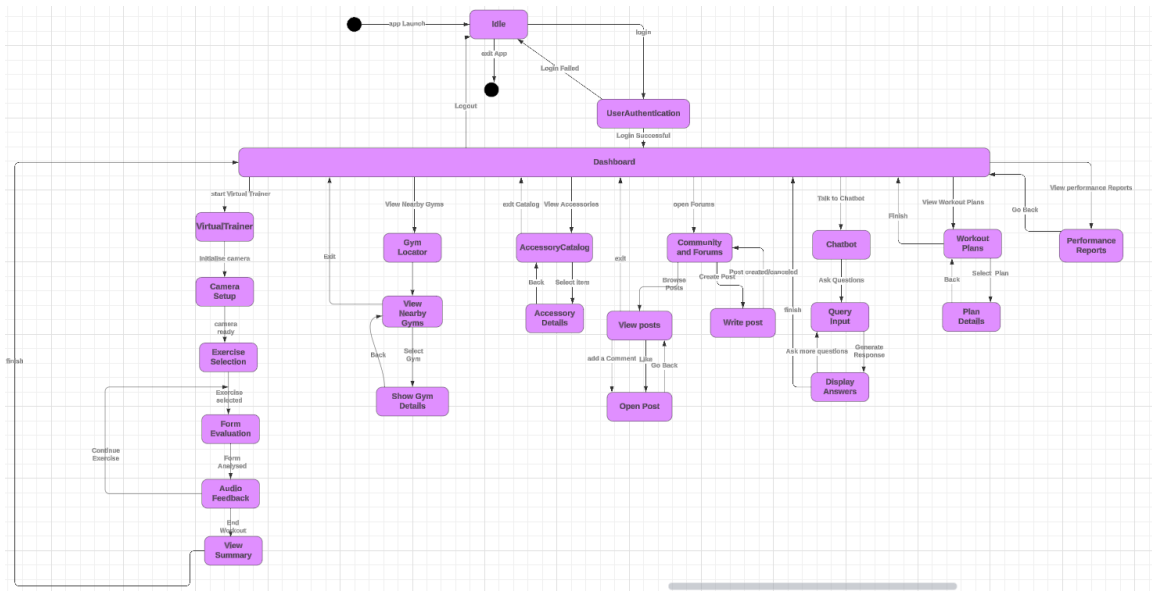


Figure 13 state transition diagram

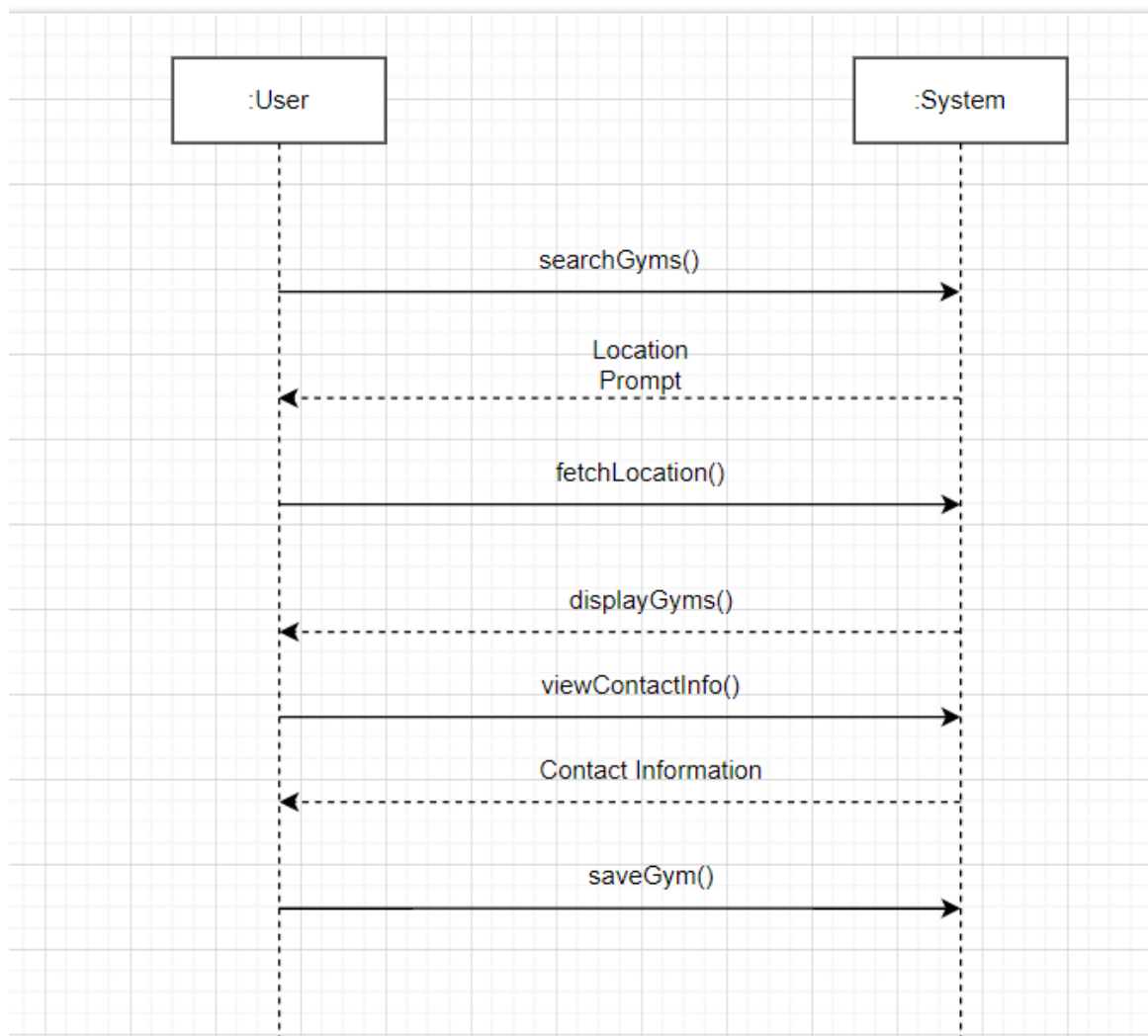
System Sequence Diagrams**1) View Accessories :**

Figure 14 System Sequence Diagram : View accessories

2) Community and Forums :

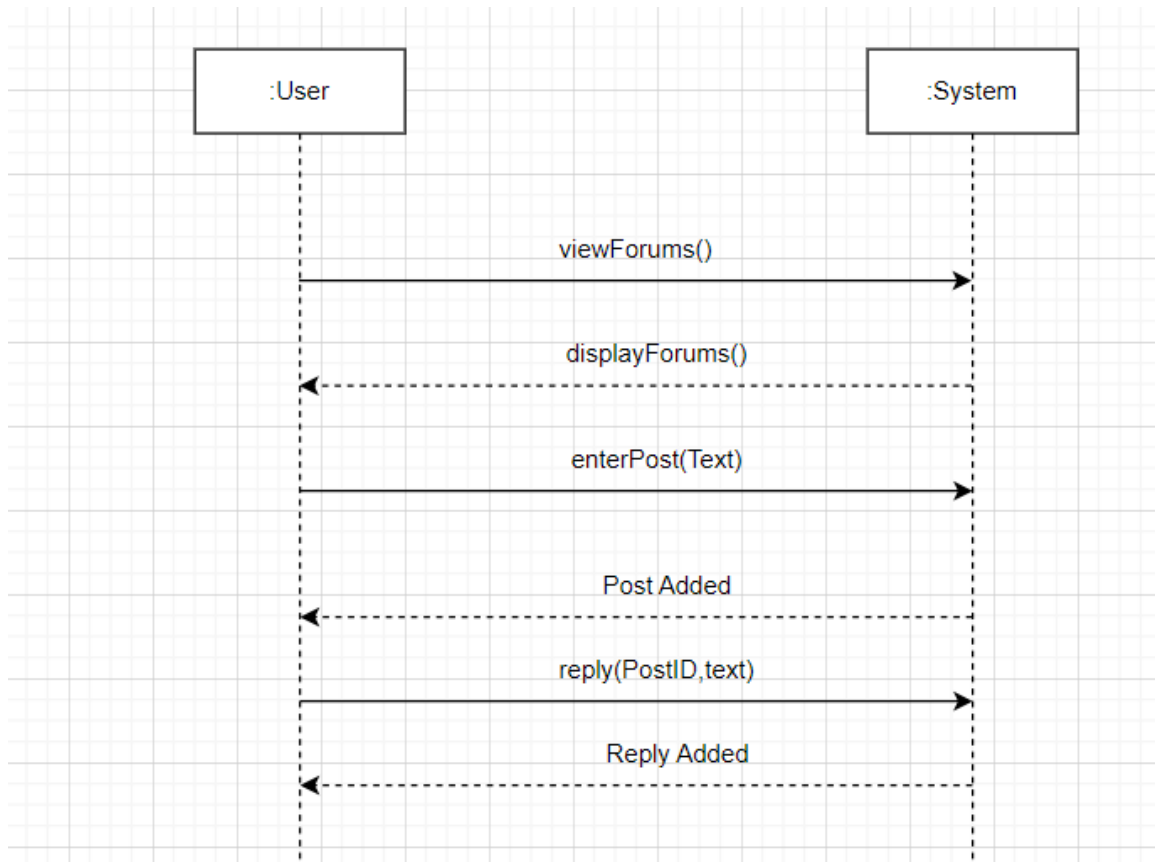


Figure 15 System Sequence Diagram : community forums

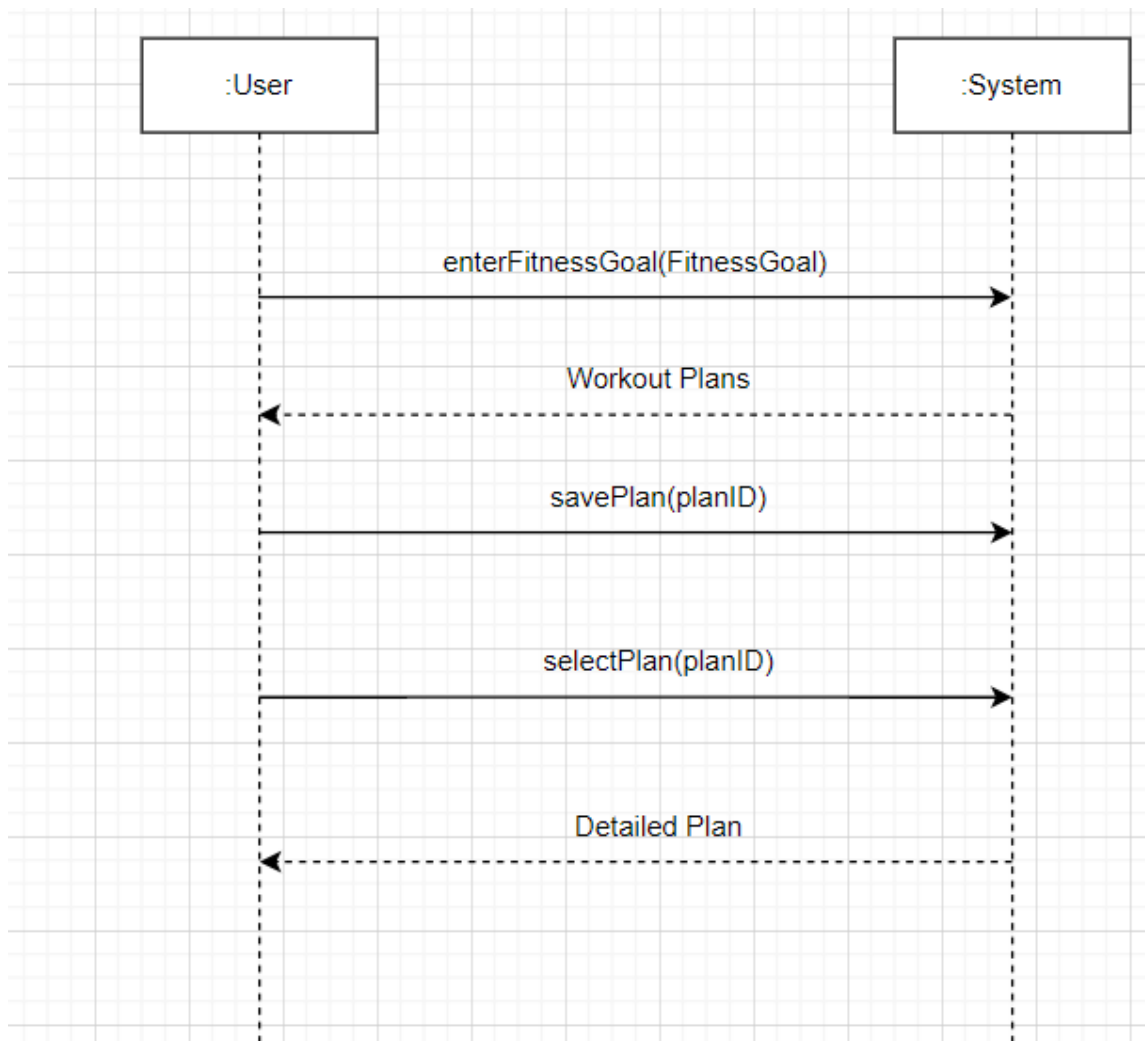
3) Personalized Workout :

Figure 16 System Sequence Diagram : Personalised Workout

4) Real-Time Workout Analysis :

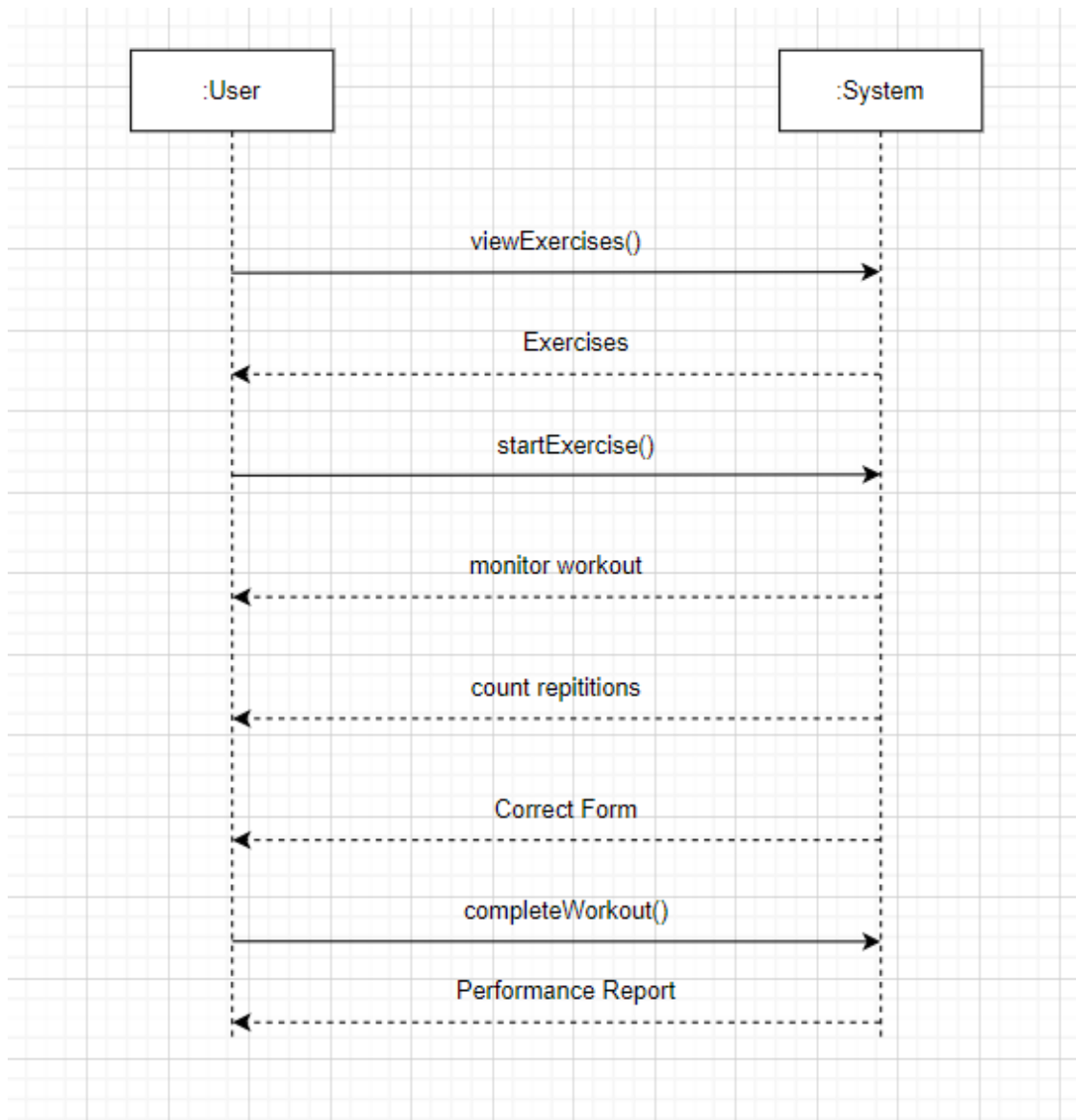
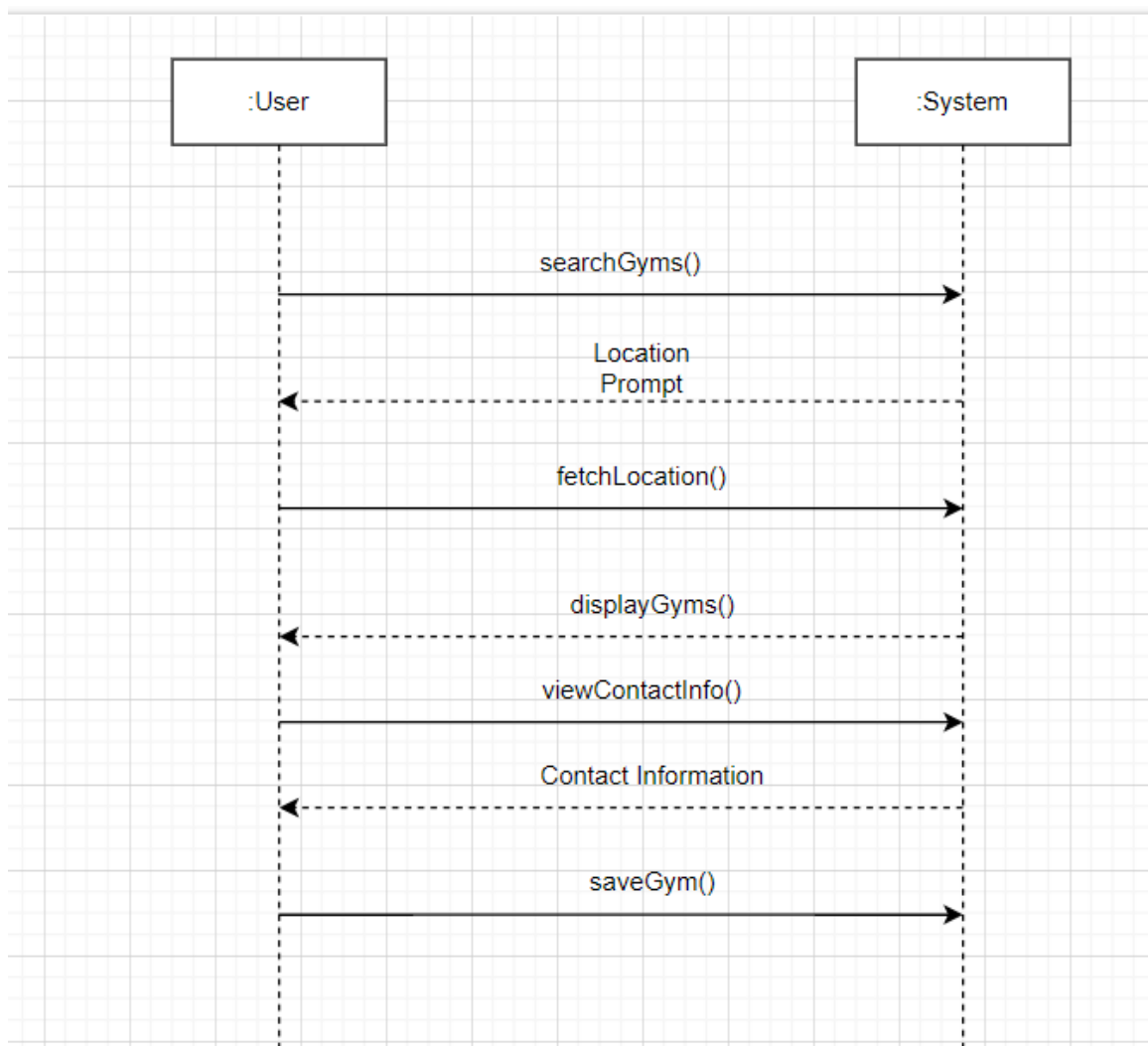


Figure 17 System Sequence Diagram : Real time Workout Analysis

5) Search Gyms :

*Figure 18 System Sequence Diagram : View Nearby Gyms*

3.3 Data Design

Our application contains multiple modules, most of which work on different types of data. The data stored by the application as a whole ranges all the way from user details to their workout analysis reports. To store our data we have opted for a Non Relational DBMS, **MongoDB**. This will help us store all data in key value pairs.

The main data sources of our application are:

1. User details
2. Workout Assessment reports
3. Accessory and store details
4. Nearby gym details
5. Workout plans
6. Community and posts

All of these will be stored in identifiable key-value pairs in separate mongoDB collections.

Data processing of each data source is explained below:

1. User details processed for user record maintenance and authorization
2. Workout Assessment reports processed for workout feedback and progress reports over time.
3. Accessory and store details for easy and efficient accessory search.
4. Nearby gym details for efficient gym search and selection.
5. Community and posts data stored and processed to manage post history and interactions.

Databases created will be:

1. Users
2. Exercises
3. WorkoutAnalysis
4. Gyms
5. AccessoryStores
6. Accessories
7. Posts
8. Comments
9. PerformanceReports
10. Progress Reports
11. Chatbot History
12. Workout

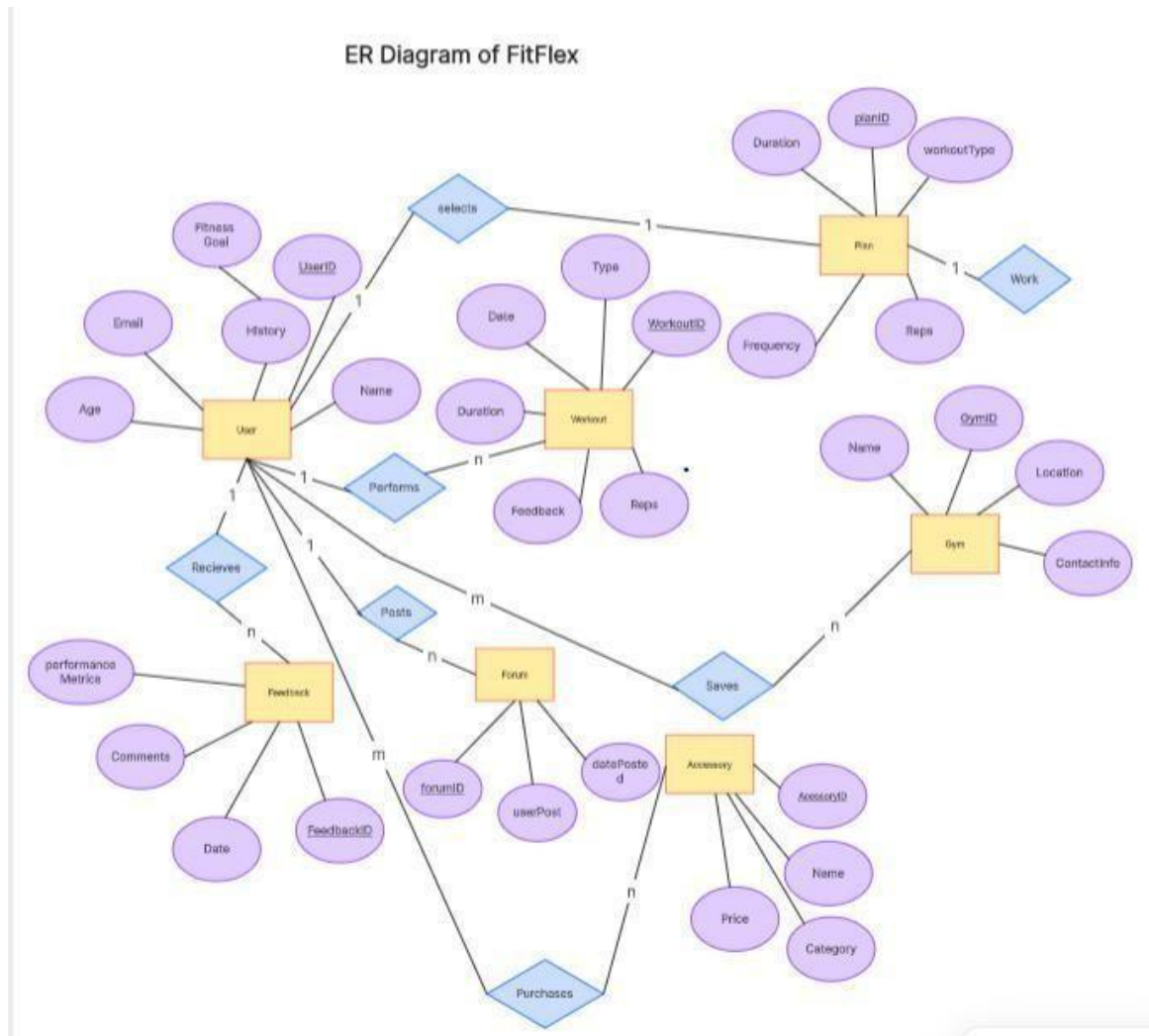
ERD

Figure 19 Entity Relationship Diagram

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- [1] A Kolyshkin and S Nazarovs. Stability of slowly diverging flows in shallow water.
Mathematical Modeling and Analysis, 2007.